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Foundations, Notation, and Mapping

The Hodge Complex Standard Model

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Substrate → Hodge Complex → Spectral Midpoint → Processor → Output → Physics & Cosmology

The framework proceeds from a single discrete substrate to observable physics.

LEVEL 0

Substrate

Periodic 8x8 torus
N = 64 sites, L = 8

LEVEL 1

Hodge Complex

$\Omega_0 + \Omega_1 + \Omega_2$
 $D = d + d^*$, dim = 256

LEVEL 2

Spectral Midpoint

Self-paired $\lambda = 4$
14 modes = 1 SM generation

LEVEL 3

Processor

$T = I - P_C$
F/B split, five-layer loop

LEVEL 4

Output Structures

Charge theorem
Mass map (0 free params)

LEVEL 5-6

Physics & Cosmology

CKM/PMNS, Higgs, Gravity
Dark energy, vacuum stability

*Physics is what the machine cannot cancel.
The Hodge complex is the machine.*

Three Discrete Inputs

- Standard Model content
- Physical identifications
- Identification thesis

One Dimensionful Anchor

- Planck mass M_P

Everything else is derived.

Fewer Anchors than SM

- Not zero anchors
- But strictly fewer empirical inputs than the Standard Model

The Periodic 8 x 8 Torus

$$\begin{aligned} \text{Lambda} &= Z8 \times Z8 \\ N &= d \cdot 2^d = 64 \quad L = 8 \end{aligned}$$

64 sites · periodic BCs · primary, not a discretization

Two Independent Anchors + One Derived

Anchor 1 Photon helicity $\rightarrow d = 4$

Anchor 2 Equal entropy spacing $\rightarrow L = 8$

Anchor 3 Kernel topology $\rightarrow \dim \ker D = 4$

Discrete 5-Point Laplacian & Spectrum

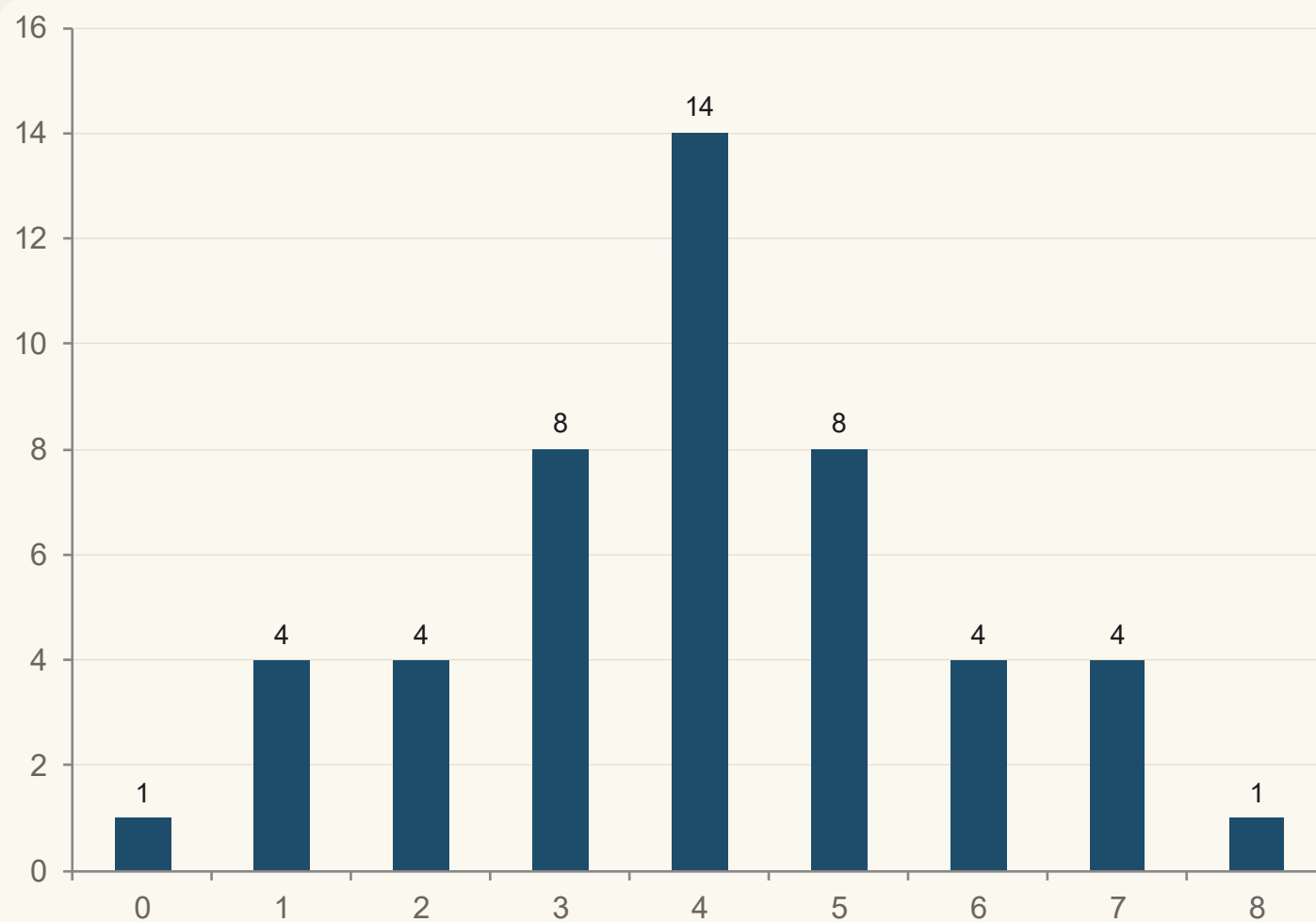
$$\text{lambda}(n1, n2) = 4 - 2 \cos(\pi n1 / 4) - 2 \cos(\pi n2 / 4) \quad \cdot \quad \text{Mirror-symmetric under } \text{lambda} \leftrightarrow 8 - \text{lambda}$$

Self-paired eigenvalue $\text{lambda} = 4$ has multiplicity $2(L - 1) = 14 \rightarrow$ matches one Standard Model generation.

Size relation: $N = d \cdot 2^d$. At $d = 4$, $N = 64 = 8^2$ (only d in $1 \dots 8$ making $d \cdot 2^d$ a perfect square).

Shannon entropies of the three D4 orbits: 6.0, 5.5, 5.0 bits — equal spacing 0.5 bits (unique for $L = 8$).

Eigenvalue spectrum of the discrete Laplacian on $Z_8 \times Z_8$



Self-Paired Midpoint

$\lambda = 4$
multiplicity = 14

Why 14 matters

The 14 modes at $\lambda = 4$ cannot cancel with any mirror partner. They are the residue that survives — identified with one Standard Model generation.

Differential forms on the torus · $D = d + d^*$

Omega0

0-forms

dim = 64

Bosonic

Functions on the 64 sites

Omega1

1-forms

dim = 128

Fermionic

Fields on 128 directed edges

Omega2

2-forms

dim = 64

Bosonic

Fields on 64 plaquettes

Key Theorems

D^2 on $\Omega_k = \Delta$ exactly on each form degree (no fermion doubling).

$\ker D$ isomorphic to de Rham cohomology: $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$.

Signature on $\ker D$ is (3,1) — Lorentzian spacetime emerges from the Hodge star.

$\mathbb{Z}_2$ grading (even = bosonic, odd = fermionic) is the origin of the spin-statistics connection.

The 14 modes at $\lambda = 4$ and their D4 orbits

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Size 2 · Support 64 = N

Entropy: 6.0 bits

$(0,4), (4,0)$

01

Size 8 · Support 48 = V

Entropy: 5.5 bits

$(1,3) \dots (7,5)$

02

Size 4 · Support 32 = N/2

Entropy: 5.0 bits

$(2,2), (2,6), (6,2), (6,6)$

Hodge Lift to the 56-Dimensional Sector

$14 \Omega_0 + 28 \Omega_1 + 14 \Omega_2 = 56$

This 56-dimensional sector is the arena of the physical world.

D4 content: $9A_1 + 5A_2 + 7B_1 + 7B_2 + 14E$

$T = I - P_C$ · Rank-10 orthogonal projection on the 14-mode multiplet

Fermion / Boson Split

$$|F| = d(d+1)/2 = 10$$

$$|B| = d = 4$$

$$(|F| - |B|)/|B| = (d-1)/2 = 3/2$$

$$|F|/|B| = (d+1)/2 = 5/2$$

Winding Classes

Class A: $\omega = 2\sqrt{2}$ (3 modes)

Class B: $\omega = 6\sqrt{2}$ (7 modes)

Class C: $\omega = 0$ (4 modes)

$$\omega_B / \omega_A = 3 = d - 1 \text{ exact}$$

Landauer Channel · Entropy reduction $\Delta S = \log(56/40) = \log(7/5) \approx 0.336$ nats saturates the deterministic Landauer bound.

The processor is a Maxwell demon operating at the exact deterministic limit. Loop-pass duration $T_{DME} = 1/(2\gamma)$.

Charge Theorem

$$q_3 = F(n_1, \sigma)$$

Bijection from 14 realized pairs
to the five SM charge eigenvalues:
 $\{-3, -1, 0, +2, +3\}$

Three sigma-lines carry charges
 $0, +5, -8 \rightarrow \text{sum} = -1e$

Charge density $\rho(x, y)$ is
exactly 4×4 periodic on the torus.

Mass Map

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)i)$$

$$A_{\text{mass}} = 224$$

$$B_{\text{mass}} \approx 0.5412$$

Coefficient $(d-1)/2 = 3/2$
derived twice:

- F/B split ratio
- DNLS Hartree orbit trace

RMS 0.0449 dex ($\sim 11\%$)
with zero free parameters.

CKM Matrix

$$\sin \theta_{12} = \pi / 14 \approx 0.2244$$

Wolfenstein:

$$\lambda = \pi/14$$

$$A = d/(d+1) \cdot (1 + \delta_{\text{edge}})$$

$$\sqrt{\rho^2 + \eta^2} = 5/12$$

Hierarchical magnitudes:

$$|V_{us}| = \lambda$$

$$|V_{cb}| = A \lambda^2$$

$$|V_{ub}| = A \lambda^3 \sqrt{\dots}$$

PMNS Matrix

Neutrino N/4 spectrum:
(60, 60, 58)

$$\theta_{12} \approx 34.2^\circ$$

$$\theta_{23} \approx 47.9^\circ$$

$$\theta_{13} \approx 8.5^\circ$$

$$\delta_{\text{CP}} \approx 197^\circ$$

Near-degenerate labels →
large mixing (vs hierarchical
quark sector).

Higgs Sector

Top Yukawa:

$$y_t^2 = 1 - (4/3) \delta_{\text{edge}}$$

$$4/3 = |O_2| / (d - 1)$$

$$m_t = 172.688 \text{ GeV}$$

(matches PDG at 0.0053σ)

$$\lambda_{\text{tree}} = \ln(kL) / [(d-1)\pi^2]$$

Bridge identity:

$$\lambda_{\text{tree}} = \gamma^2/d \cdot (1 - \dots)$$

$$v_{\text{EW}} = M_P \exp(-k L_{\text{phys}})$$

Gravity

Graviton = self-paired

B1 + B2 curvature mode

(108 modes, exactly massless)

$$\begin{aligned} G_N &= (d+1)^2 / (8 \pi M_P^2) \\ &= 25 / (8 \pi M_P^2) \end{aligned}$$

Factor 25 = $(\dim V_{B1})^2$

arises as the trace of the
bilinear form on the hidden
B1 isotypic sector.

kL Crossover

$kL_{\text{bare}} = (d-1)N/(d+1) = 38.4$ exact
 $kL_{\text{phys}} = 38.442527$
 Matches $\ln(M_P / v_{\text{EW}})$ to 0.004%

Dark Energy

$\rho_{\text{DE}} = (d+1)/2 \cdot m_{\nu}^4$
 $= 2.4809 \times 10^{-47} \text{ GeV}^4$
 Matches observation to 0.76%
 $w = -1$ exact (topological Casimir)

Vacuum Stability

$\gamma = c_2 / l(2) \approx 0.70284$
 $l(2)$ from AGM(1, $\sqrt{2}$) & $\eta(i)$
 $S''(0)/S_0 = -2\gamma < 0$
 $D = 4$ is a strict local maximum

Quantity	Derived Expression
Top-Yukawa coefficient	$4/3 = O_2 /(d-1)$
Tree-level quartic λ_{tree}	$\ln(kL) / [(d-1)\pi^2]$
Cabibbo angle $\sin \theta_{12}$	$\pi/14$
Neutrino $N/4$ spectrum	$(60, 60, 58)$
Mass-map coefficient	$(d-1)/2 = 3/2$
Loop-pass duration T_{DME}	$1/(2 \gamma)$
Newton constant G_N	$(d+1)^2 / (8 \pi M_P^2)$
Dark-energy coefficient	$(d+1)/2$

The continuum emerges from the substrate — not the other way around.

Physical vs lattice scale

Lattice spacing is a structural count, not a physical length. $\epsilon = \text{lattice}/\text{physical scale} \rightarrow 0$ in the scaling limit.

Isotropic dispersion

$\lambda(n) = (\pi^2/16)(n_1^2 + n_2^2) + O(n^4)$. Quartic anisotropy suppressed by ϵ^2 in the IR.

Cohomological content

Physical observables live in cohomology of the Hodge complex — topological invariants independent of lattice spacing.

Emergent $SO(3,1)$

Lorentz group acts on $\ker D$ via intersection form & Hodge inner product. Exact continuous symmetry, no lattice remnant.

Conclusion

The Standard Model is what the substrate looks like when you observe it.

*The Hodge complex is the machine.
Physics is what the machine cannot cancel.*

Full paper stack & verification scripts:
github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM-